

Data Flow Diagram

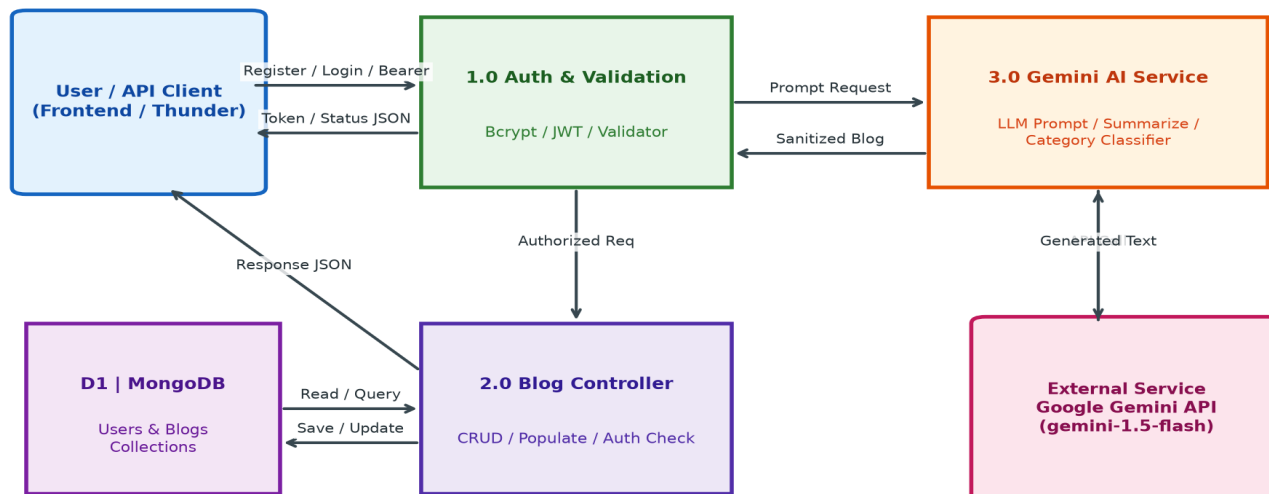
Date:	3 October 2026	Team ID:	UNM1611SHARVESHARAM-SWTID-2026-8110
Project Name:	AI BlogNest API	Maximum Marks:	2 Marks

Data Flow Diagram (DFD) Specification:

A Data Flow Diagram (DFD) maps out the flow of information through AI BlogNest API, detailing interactions between external entities, backend processes, and persistent data stores. Use the standard symbol legend below as reference.

Symbol Shape	Element Name	Description	Project Implementation in AI BlogNest
Oval / Rounded	External Entity	A person, system, or third-party service that sends or receives data.	• User / API Client (Thunder Client / Web UI) • Google Gemini Generative AI Cloud API
Rectangle with Number	Process	An activity or business logic that transforms incoming data into outgoing data.	• 1.0 Auth & Validation (Bcrypt / JWT) • 2.0 Blog Controller (CRUD & Ownership) • 3.0 Gemini AI Service (Prompt / Sanitize)
Rectangle (Solid / Open)	Data Store	A persistent repository where structured data is stored for later use.	• D1: MongoDB Users Collection • D2: MongoDB Blogs Collection
Labeled Arrow	Data Flow	Directed movement of data labeled with specific data fields.	• Credentials, JWT tokens, Prompt topics, Sanitized blog text, Mongoose documents

Level-1 Data Flow Diagram (DFD) — AI BlogNest API



Data Flow Process Descriptions:

Step	Source -> Target	Data Transferred	Transformation & Security Logic
1	Client -> Auth Process	name, email, password	express-validator checks format; bcrypt hashes password (salt=10); user saved to D1.
2	Auth -> Client	HMAC-SHA256 JWT Token	Payload contains userId; signed with secret; valid for 7 days.
3	Client -> AI Service	Bearer Token + topic + category	authMiddleware validates token; prompt constructed with 100-150 word limit constraint.
4	AI Service -> Gemini API	Engineered Prompt string	Secure API key dispatch to gemini-1.5-flash endpoint.
5	Gemini API -> AI Service	Raw LLM Text Response	Regex sanitizes markdown/asterisks; keyword engine classifies domain category.
6	AI Service -> MongoDB (D2)	Blog document object	Persists title, content, category, author ObjectId, authorName, timestamps.